Vito Marco Rubino

AI Engineer & Data Scientist

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Experience

Engineering Ingegneria Informatica S.p.A. Jul 2026 – Present
Data Science Advisor Rome, IT

- Built an agentic AI assistant and RAG pipeline for intra- and cross-document search and form-filling automation across **7,000+ vector chunks**, using FastAPI, PGVector, and AWS (Kubernetes, S3) to serve **17,000+ users**.
- Sped up document translation **from 15 minutes to 0.5 seconds** using a CPU-optimized model.
- Developed AI Smart Agents for anomaly detection, multi-source correlation and behavior prediction.

Officina Futuro Fondazione W-Group ETS Sep 2022 – Apr 2025
Coach Maker Bari, IT

- Mentored **80 girls** over **110 hours** of '*Girls Code It Better*' workshops, leading the development of **3** Cloud, Sustainability, and AI prototypes showcased at the national conference.

Education

University of Bari Aldo Moro Sep 2024 – Jun 2026
M.S. in Data Science, 110/110 Summa Cum Laude

- Thesis: "Hallucinations Detection in LLMs Integrating RAG across Heterogeneous Domains"

University of Bari Aldo Moro Sep 2021 – Jun 2024
B.S. in Computer Science and Technologies for Software Production, 110/110 Summa Cum Laude

- Thesis: "CNNs and Attention Mechanisms for Parkinson's Diagnosis and Speech Deficit Detection"

Projects

- **CNNs and Attention Mechanisms for Parkinson's Diagnosis**
  Reached 96.6% accuracy in early Parkinson's diagnosis from voice recordings by using **CNN** models and applying Grad-CAM for interpretability. Awarded the '**Best Thesis Award**' and published in **Springer Nature**.
- **Airline Passenger Satisfaction Analysis**
  Defined customer retention strategies by segmenting 5 distinct passenger profiles, identified through the benchmarking of 17 clustering models and rule mining techniques including Apriori and FP-Growth.
- **Amazon Electronics Reviews Analysis**
  Identified sentiment drivers, market hubs, and validated a recommender system with an RMSE of 1.12 by processing 20 million Amazon reviews using **NLP**, Word2Vec, PageRank, and ALS on a distributed Spark architecture.
- **Wind Turbine Maintenance Optimization**
  Minimized maintenance costs for a wind farm by managing 70 binary variables and 17 constraints through a BILP model solved using the Branch-and-Cut algorithm.

Skills

- **Data Science & AI:** Machine Learning, Deep Learning, PyTorch, Natural Language Processing, Cluster Analysis, Computer Vision, YOLO, Data Mining
- **Big Data & Database:** MySQL, MongoDB, Apache Spark
- **Programming:** Python, Java, R, Scala, C++
- **Tools and Methodologies:** Scrum, LaTeX, Git, Android Development, Software Testing (JUnit, TDD)

Languages

- **English:** Professional working proficiency
  - **Cambridge English** - B2 First (FCE)
- **Italian:** Native proficiency

Awards

- **Best Thesis Award** – Parkinson's Diagnosis Jan 2025
  - Awarded by SERLAB and SER&P for research excellence in Parkinson's diagnosis, recognized for the innovative application of CNNs and Attention Mechanisms.

Publications

[1] R. Capolongo, V. Gattulli, D. Impedovo, G. Marangi, L. Musti, and V. M. Rubino, "Explainable and real-time road pothole analysis using object detection models", en, in *Proceedings of the 6th National Conference on Artificial Intelligence (Ital-IA 2026)*, Organized by CINI, Rome, Italy, Jun. 2026.

[2] L. Simone, M. G. Camporeale, V. M. Rubino, V. Gervasi, and G. Dimauro, "Interpretable Early Detection of Parkinson's Disease Through Speech Analysis", en, in *Artificial Intelligence in Medicine*, R. Bellazzi, J. M. Juarez Herrero, L. Sacchi, and B. Zupan, Eds., vol. 15735, Cham: Springer Nature Switzerland, 2025, pp. 373–378, ISBN: 9783031958403 9783031958410.

Extracurricular Activities

- Computer Science and Mathematics Tutor 2022 – Present
  Tutored high school and university students in preparing for mathematics and computer science exams.

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